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# Load Packages -----
library(shiny)
library(tidyverse)
library(readxl)
library(tigris)
library(leaflet)
library(sf)
library(plotly)
library(shinydashboard)
library(echarts4r)
library(timevis) # added for timeline

options(tigris_use_cache = TRUE)

# Data -----
getwd()
# Import data
labor_data <- read_csv("Data/WB Output/VA_data.csv")
# Create VA data
VA_data <- labor_data %>%
  filter(STATE_NAME == "Virginia")

# Prepare VA counties spatial data
va_counties <- counties(state = "VA", cb = TRUE, class = "sf") %>%
  mutate(FIPS = GEOID) %>%
  st_transform(crs = 4326)

# Format FIPS codes consistently
VA_data <- VA_data %>%
  mutate(COUNTY_FIPS_CODE = str_pad(as.character(COUNTY_FIPS_CODE), width = 5,
pad = "0"))

# Join spatial and attribute data
map_data_all <- va_counties %>%
  left_join(VA_data, by = c("GEOID" = "COUNTY_FIPS_CODE"))

VDOE_data <- read_csv("Data/WB+VDOE copy.csv")
total_by_date <- VDOE_data %>%
  group_by(date) %>%
  summarise(Total_Count = sum(Total_Count, na.rm = TRUE)) %>%
  ungroup()

# Policy data moved outside UI -----
policy_data <- data.frame(

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id = 1:13,
content = c(
  "2018 Reimbursement Rate Increase",
  "2020 CCDF Funding Allocation",
  "2020 CCDF Transition Plan",
  "2021 COVID-19 Relief",
  "2021 House Bill 2206",
  "2021 House Bill 1800",
  "2022 House Bill 30",
  "2022 Supplier Correspondence on Copays",
  "2022 Supplier Correspondence on Revisions to Provider Payment Rates",
  "2023 Correspondence on Update in Paid Childcare Absences",
  "2024 HB419",
  "2024 Provider Correspondence on Waitlists Being Reinstated",
  "2024 HB627"
),
start = as.Date(c(
  "2018-06-01",
  "2020-10-01",
  "2020-11-24",
  "2021-02-22",
  "2021-01-28",
  "2021-01-01",
  "2022-01-01",
  "2022-12-06",
  "2022-09-29",
  "2023-09-20",
  "2024-02-07",
  "2024-06-04",
  "2024-01-10"
)),
summary = c(
  "In 2018, Virginia increased reimbursement rates for licensed providers to align more closely with the 70th percentile of the market rate survey.",
  "In 2020, a report was released explaining funding allocation within CCDF.",
  "The CCDF was transitioned from the Virginia Department of Social Services (VDSS) to the Virginia Department of Education (VDOE). The plan included a phased shift to bringing all aspects of subsidies to the VDOE. It also brought up the shift to Virginia Quality Initiatives (VQI) and how it will be implemented in phases, along with impact of COVID-19 on childcare and updates on supporting CCDF through COVID-19.",
  "In 2021 the CCDF was given funds under the Coronavirus Response and Relief Supplemental appropriations Act (CRRSA). The goals to utilize these funds was to increase the supply of childcare, increase resources for providers and families, expand requirements for

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subsidies, and to increase access to childcare in underserved communities. Copayments were eliminated and other eligibility requirements would soon be changing as well.",

"A bill temporarily expanding subsidy program to those who are actively looking for work and raising the qualifying threshold to those who's income is below 85% of the state median income.",

"Provided 9.6 Million Dollars to CCDF to expand the subsidy program.",

"This bill maximized the use of federal CCDF funds to eliminate the waitlist for subsidies. It also continued the 85% of median income eligibility to the 2022-2024 biennium.",

"This correspondence informed subsidy providers that copays will be reinstated January 1 2023, and brought up that with it would be a new copayment scale.",

"This correspondence informed subsidy providers that payment rates would be changing to fit a new cost estimation model, giving a majority of centers increased rates. It also mentioned new copayment scales, along with subsidy providers now being paid for 15 days of planned holidays.",

"This correspondence effectively changed the number of allotted absence days still paid to the provider from 36 to 60 days a year.",

"This bill amended the codes that handled funding for childcare, teducators indecisive, and nonreverting funds to capture unspent funding balances.",

"This correspondence explained that due to need that does not meet supply subsidy waitlists will be reinstated throughout the state.",

"This bill was to provide subsidies to childcare employees regardless of income status, in hopes of incentivizing working in childcare. It passed, but then died in the house."

),

link = c(

"timeline/2018-06-MR-Survey.qmd.pdf",

"timeline/2020-10-01-RD367.pdf",

"timeline/2020-11-24-CCDF-transition-plan.pdf",

"timeline/2021-02-22-Covid-Relief.pdf",

"timeline/2021-01-28-HB2206.pdf",

"timeline/2021-HB1800.pdf",

"timeline/2022-HB30.pdf",

"timeline/2022-12-6-ProvLet-Copays.pdf",

"timeline/2022-09-29-Corresp...ider-payment-rates.pdf",

"timeline/2023-09-20-paidabsenceincrease.pdf",

"timeline/2024-02-07-HB419.pdf",

"timeline/2024-06-04-waitlistvendorsletter.pdf",

"timeline/2024-01-10-subsidysexpandtoCCworkers.pdf"

),

stringsAsFactors = FALSE

)

policy_data\$start <- as.character(policy_data\$start)

UI -----

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ui <- navbarPage(
  title = "Childcare Analysis Project",

  header = tagList(
    tags$head(
      tags$style(HTML("
        body {
          background-color: #ffffff;
          font-family: 'Times New Roman', serif;
        }
        .navbar {
          background-color: #2e708c;
        }
        .navbar-default .navbar-nav > li > a {
          color: white;
        }
        .navbar-default .navbar-brand {
          color: white;
          font-family: 'Times New Roman', serif;
          font-weight: bold;
          font-size: 20px;
        }
        .tab-content {
          background-color: #d3ebf5;
        }
        h2 {
          color: #333333;
          font-family: 'Times New Roman', serif;
          style = font-size:38px; font-weight:bold; margin-bottom:10px;
        }
        p {
          color: #333333;
          font-family: 'Times New Roman', serif;
          style =font-size:20px
        }
      ")))
  ),

  tabPanel("Background",
    fluidPage(
      div(
        style = "text-align: center;",
        h2("Exploring the Impact of Childcare on Labor Force Outcomes",

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    style = "font-size:38px; font-weight:bold; margin-bottom:10px;"),
    h3("Data Science for the Public Good",
      style = "font-size:17px; font-weight:bold; margin-top:0px; margin-bottom:5px;"),
    h3("Virginia Tech 2025",
      style = "font-size:17px; font-weight:bold; margin-top:0px;")
  )

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    ,
    fluidRow(
      column(
        width = 6,
        h4("Background:",
          style = "font-weight:bold"),
        p("Childcare acts as a foundational pillar for economic stability and workforce engagement; however, the rising cost of care has become a major barrier to employment. According to the Center for American Progress, the lack of affordable childcare is one of the most significant obstacles to sustained employment, leading to parents reducing hours, turning down opportunities, or exiting the labor force altogether. The burden associated with childcare and the workforce falls disproportionately on mothers, who are more likely to bear caregiving responsibilities. When women are unable to find proper childcare, they are less likely to be employed, reducing household income with potentially long-term career setbacks."),
        p("Moreover, the affordability and availability of childcare vary widely across different regions nationally and within the state of Virginia. Families living in rural areas or economically disadvantaged regions face additional barriers, including fewer childcare providers. These disparities exacerbate labor market inequalities and deepen the economic precarity of working families."),
        p("Childcare subsidies help low and moderate-income families afford the cost of childcare, enabling parents to work, look for work, attend school, or participate in vocational training. The primary federal program providing this funding is the Child Care and Development Fund (CCDF), which in the state of Virginia has been administered and maintained by the Virginia Department of Education since 2020, priorly being managed by the Virginia Department of Social Services.")
      ),
      column(
        width = 6,
        h4(" "),
        p("Since 2018, there have been many changes made to the eligibility of subsidies, particularly during and after the COVID-19 pandemic, both nationally and at the state level. In response to the crisis, $9.6 billion in federal funding was provided through the Coronavirus Response and Relief Supplemental Appropriations Act. Enabling the state of Virginia to temporarily expand eligibility and serve more families. The family income threshold was raised to 85% of the federal poverty line, and the eligibility requirements were expanded to parents actively searching for work to qualify for subsidies; an important change that supported families facing job loss or economic instability associated with the pandemic. During this time,")
      )
    )

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copayments and waitlists for subsidies were eliminated. These two supportive measures have been rolled back in 2023 and 2024, respectively, due to budget constraints; however, the pandemic-era guideline for income requirements and job-seeking parents has remained as eligibility requirements for childcare assistance."),

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      h4("Research Questions:",
        style = "font-weight:bold"),
      tags$ul(
        tags$li("How do childcare prices differ across the state of Virginia, specifically
between rural and urban areas?"),
        tags$li("What is the relationship between labor outcomes and the usage of
childcare, and how do subsidy usage, gender, and locality play a role in said relationship?"),
        tags$li("How do changes in Virginia state policies impact the program participation,
and how does this demand differ across the state?")
      )
    ),
    br()
  )
),
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tabPanel("Methodology",
  fluidPage(
    h2("Methodology"),
    p("Details of your methodology."),
    p("Details of our methodology.")
  )
),
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```
tabPanel("Understanding Policy",
  fluidPage(
    h2("Timeline"),
    titlePanel("Virginia Childcare Policy Timeline"),
    fluidRow(
      column(8, timevisOutput("timeline")),
      column(4, uiOutput("policyDetails"))
    ),
    p("Click a policy event on the timeline above to see where it falls—highlighted by a
yellow line—on the chart of total children receiving subsidies from 2018 to 2025."),
    br(),
    fluidRow(
      column(12, echarts4rOutput("line_plot"))
    )
  )
),
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tabPanel("Text Mining Info",
  fluidPage(
    h2("Text Mining Info"),
    p("Explain text mining methods or results."),
    p("Explaining text mining methods or results.")
  )
),

tabPanel("Maps",
  fluidPage(
    h2("Maps"),
    p("Insert your map visualizations."),
    h2("Virginia Childcare Costs as a Percent of Annual Family Income"),
    sidebarLayout(
      sidebarPanel(
        selectInput("STUDYYEAR", "Select Year", choices = 2009:2022),
        selectInput("percentile", "Select Pricing Point:", choices = c("50th Percentile"="M",
"75th Percentile"="_75")),
        selectInput("age_group", "Select Age Group:", choices = c(
          "0 to 5 Months" = "Bto5",
          "6 to 11 Months" = "6to11",
          "12 to 17 Months" = "12to17",
          "18 to 23 Months" = "18to23",
          "24 to 29 Months" = "24to29",
          "30 to 35 Months" = "30to35",
          "36 to 41 Months" = "36to41",
          "42 to 47 Months" = "42to47",
          "48 to 53 Months" = "48to53",
          "54 Months to School Age" = "54toSA",
          "School Age" = "SA"
        )),
        selectInput("provider", "Select Provider Type:", choices = c("Center Care"="C",
"Home Care"="FCC"))
      ),
      mainPanel(
        leafletOutput("childcareMap", height = "600px")
      )
    )
  ),

tabPanel("Research Implications",
  fluidPage(

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        h2("Implications"),
        p("Describe your models or predictions."),
        p("Describing the models or predictions.")
    )
),
tabPanel("About Us",
    fluidPage(
        div(
            style = "display: flex; justify-content: center;",
            div(
                style = "text-align: center; max-width: 900px;",

                h2("About Us", style = "font-size:38px; font-weight:bold; margin-bottom:10px;"),

                p(HTML('Data Science for the Public Good (DSPG) Young Scholars program is an
intensive summer program hosted by the Virginia Tech Department of Agricultural and Applied
Economics and the Virginia Cooperative Extension Service. This summer marks five years of
the immersive ten-week program enabling students from a variety of colleges and universities to
have hands-on educational experience working with data analysis, statistics, R coding, and the
social sciences. These skills are applied to real-world social issues via team-based research
projects to enable students to utilize data for real-world issues such as policy, industry,
academic research, and community engagement. For more information on the program, other
teams\' research, and information on how to apply, visit the official <a
href="https://aaec.vt.edu/academics/undergraduate/dspg.html" target="_blank">VT DSPG
site</a>.')),

                h3("Undergraduate Fellows", style ="font-size:32px; font-weight:bold;
margin-top:0px"),
                fluidRow(
                    column(
                        width = 4,
                        div(style = "text-align:center;",
                            img(src = "images/grace.jpg", height = "200px", style = "border-radius: 50%;
margin-bottom: 10px;"),
                            br(),
                            tags$a("Grace Mullins", href = "https://www.linkedin.com/in/grace-eliz-mullins/",
target = "_blank",
                                style = "font-size:16px; font-weight:bold; text-decoration:underline;
color:#0072B2;"),
                            p("Berea College '27", style = "font-size:14px; color:#333333; margin-top:0px;
margin-bottom:0px;"),
                            p("Sociology", style = "font-size:14px; color:#333333; margin-top:0px;") )
                        ),

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column(
  width = 4,
  div(style = "text-align:center;",
    img(src = "images/yin.jpeg", height = "200px", style = "border-radius: 50%;
margin-bottom: 10px;"),
    br(),
    tags$a("Yin Kyay Wai", href = "https://www.linkedin.com/in/yin-kyay-wai/", target
= "_blank",
      style = "font-size:16px; font-weight:bold; text-decoration:underline;
color:#0072B2;"),
    p("Berea College '26", style = "font-size:14px; color:#333333; margin-top:0px;
margin-bottom:0px;"),
    p("Computer Science", style = "font-size:14px; color:#333333; margin-top:0px;")
  )
),

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column(
  width = 4,
  div(style = "text-align:center;",
    img(src = "images/joshua.jpeg", height = "200px", style = "border-radius: 50%;
margin-bottom: 10px;"),
    br(),
    tags$a("Joshua Yoo", href =
"https://www.linkedin.com/in/joshua-yoo-6b389524a/", target = "_blank",
      style = "font-size:16px; font-weight:bold; text-decoration:underline;
color:#0072B2;"),
    p("The College of William and Mary '26",style = "font-size:14px; color:#333333;
margin-top:0px; margin-bottom:0px;"),
    p("Finance and Computational and Applied Mathematics", style =
"font-size:14px; color:#333333; margin-top:0px;")
  )
),

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h3("Graduate Fellow", style ="font-size:32px; font-weight:bold; margin-top:30px"),
fluidRow(
  column(
    width = 12,
    div(style = "text-align:center;",
      img(src = "images/xiaoyi.jpeg", height = "200px", style = "border-radius: 50%;
margin-bottom: 10px;"),
      br(),
      tags$a("Xiaoyi Zhao", href =
"https://www.linkedin.com/in/xiaoyi-zhao-32457a248/", target = "_blank",

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    ) %>%
    addLegend(
      pal = pal,
      values = df$cost_pct_income,
      title = "Cost as % of Income",
      labFormat = labelFormat(suffix = "%", transform = function(x) 100 * x)
    )
  })

# Timeline outputs
output$timeline <- renderTimevis({
  timevis(
    data = policy_data,
    options = list(selectable = TRUE)
  )
})

output$policyDetails <- renderUI({
  selected_id <- input$timeline_selected

  if (!is.null(selected_id)) {
    selected_row <- policy_data[policy_data$id == selected_id, ]

    tagList(
      h4(selected_row$content),
      p(selected_row$summary),
      tags$a(
        href = selected_row$link,
        target = "_blank",
        " View Document",
        style = "color: blue; text-decoration: underline; cursor: pointer;"
      )
    )
  } else {
    h5(" Click a timeline event to view its summary and document.")
  }
})

selected_range <- reactiveVal(NULL)

observeEvent(input$timeline_selected, {
  selected_id <- input$timeline_selected
  if (!is.null(selected_id)) {
    selected_date <- as.Date(policy_data$start[policy_data$id == selected_id])
    selected_range(c(selected_date - 30, selected_date + 30))
  }
})

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    } else {
      selected_range(NULL)
    }
  })
output$line_plot <- renderEcharts4r({
  p <- total_by_date %>%
    e_charts(date) %>%
    e_line(Total_Count, name = "Total Count") %>%
    e_tooltip(trigger = "axis") %>%
    e_x_axis(
      type = "time",
      name = "Date",
      axisLabel = list(
        formatter = htmlwidgets::JS("function (value) { return new
Date(value).toLocaleDateString(); }")
      )
    ) %>%
    e_y_axis(name = "Total Count", min = 0) %>%
    e_title("Total Count of Children on Subsidies Over Time") %>%
    e_datazoom(type = "slider", start = 0, end = 100)

# ☒ Add vertical policy lines (must convert to character)
for (d in as.Date(policy_data$start)) {
  p <- p %>%
    e_mark_line(
      data = data.frame(xAxis = as.character(d)),
      symbol = "none",
      lineStyle = list(type = "dashed", color = "gray")
    )
}

# ☒ Add shaded area if a range is selected
if (!is.null(selected_range())) {
  range <- selected_range()

  p <- p %>%
    e_mark_area(
      data = list(
        list(xAxis = as.character(range[1])),
        list(xAxis = as.character(range[2]))
      ),
      itemStyle = list(color = "rgba(255, 165, 0, 0.3)")
    ) %>%
    e_mark_line(

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      data = data.frame(xAxis = as.character(range[1])),
      symbol = "none",
      lineStyle = list(type = "solid", color = "blue", width = 2)
    ) %>%
    e_mark_line(
      data = data.frame(xAxis = as.character(range[2])),
      symbol = "none",
      lineStyle = list(type = "solid", color = "blue", width = 2)
    )
  }

  p
})
}
# Run App -----
shinyApp(ui = ui, server = server)

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